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Factors affecting efficient discharge of judicial functions: Insights from Indian courts

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ABSTRACT

The mandate of any judicial system is to pronounce quality judgements in a timely, impartial and transparent manner. The Indian judiciary is currently facing a mammoth backlog of cases leading to delayed delivery of justice. The present work develops a framework identify and prioritize factors that influence judicial performance and applies it to the case of Indian courts. The study determines twenty factors that affect an efficient delivery of judicial functions. We classify these factors into five broad categories, namely, organizational, judge-related, case-related, procedural and exogenous. We rank the categories and factors within each category using fuzzy best-worst method. Findings indicate that organizational and procedural categories are most significant, thereby, having maximum potential to improve judicial efficiency. Complexity associated with judicial procedures is the highest ranked factor, followed by lack of information technology initiatives, high complexity of cases, low judicial staff and procedural ambiguities. We recommend policy interventions based on key insights derived from the results. Focused changes in acts and codes for simplification of judicial processes are essential to enhance judicial efficiency. Judges and staff can play a more active role by adopting practices to improve case procedures. Various challenges associated with technological reforms need to be addressed for their successful implementation. Separate procedures can be outlined for complex cases and judges can prioritize these cases to reduce unnecessary delay. Courts can also recruit judicial staff in line with the requirements followed by their optimal allocations.

1. Introduction

The judicial system of any country is established on the principle of fairness. This essentially means that every individual has the right to their day in court: to have their case heard fairly, wherein the case is processed through established procedures, and judgement is delivered in an impartial manner. The timely resolution of cases is an essential requirement of a judicial system [1–3]. The courts should be capable of pronouncing decisions in predictable time frames without compromising on the relevant practices necessary for each case. In fact, delayed judgements may not justify the effort and money spent by parties seeking judicial remedies.

The Indian judiciary is known for high pendency levels and slow delivery of justice. There are over 50 million pending cases in courts across the country (according to National Judicial Data Grid on July 25, 2023). Over one-fourth of these have been pending for over five years. Less number of judges to match the judicial demand is often cited as the main reason behind pending cases. Hence, enhancing judicial strength is a frequently proposed solution to combat case backlog [4,5]. Measures to increase judges include expediting judicial appointments, streamlining the process of recruitment, adding temporary courts, and engaging retired judges on an ad-hoc basis [6,7]. However, it is possible that adding judges alone may not have the expected impact and other reforms may be needed.

The judicial systems across countries vary in terms of their organizational structure, operations, laws and processes. This may lead to differences in the influence of various factors on judicial performance; same factors may differ in terms of direction and magnitude of impact. Accordingly, same improvement measures may not provide similar results across different court systems, thereby preventing generalization or replication of reforms [8]. Policy reforms should be developed according to the context and factors affecting judicial functioning. Specific reforms that focus on the most significant factors or the factors with the maximum scope for improvement are likely to yield long-term efficiency gains. This necessitates a study to identify and analyse factors that affect the efficiency of Indian courts. Once the factors are determined, relevant

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interventions can be informed to enhance judicial performance.

The present work aims to develop a methodological framework for the identification and prioritization of factors that affect judicial efficiency. The framework is applied to the case of Indian courts. For the purpose of our study, the terms, 'judicial performance' and 'judicial efficiency' represent the quantitative performance of courts. Judicial quality is intrinsically included in judicial efficiency since courts should deliver fair judgements without unnecessary delays [9]. In fact, quality and quantity are not incompatible goals; improvements in court output may not necessarily lead to poor adjudicatory quality [10,11]. The factors relevant to the Indian context are determined through a review of literature and discussions with experts from the judicial domain. Fuzzy best-worst method (FBWM), a multi-criteria decision-making (MCDM) tool is used to prioritize the factors based on their relative importance. Key insights are derived from the results and accordingly, policy reforms are suggested to enhance judicial efficiency.

The objectives of the study are as follows:

1. To develop a methodology that enables the identification of factors that affect the efficient functioning of courts and apply that to the case of India.
2. To prioritize the factors according to their relative impact and suggest relevant policy interventions.

The rest of the paper is structured as follows: section 2 presents a review of literature on various factors that impact judicial performance. Section 3 describes the methodology adopted in the study followed by the results in section 4. Section 5 presents the discussion and policy insights derived from the study. Section 6 concludes the paper with a summary of the study and directions for future research.

2. Literature review

The review of literature comprises of studies that evaluate the impact of various factors on judicial performance. The review is divided into eights sections from section 2.1 to 2.8. The research gaps are highlighted in section 2.9.

2.1. Financial resources

Budgeting models that rely on performance tend to have an equitable distribution of resources and similar judicial efficiency in courts across the country. Allocation of financial resources that is not in proportion to the caseload can lead to differences in performance of courts within the country [12]. Peyrache & Zago [13] show that sub-optimal reallocation of resources contributes nearly one-third of the total inefficiency in Italian courts. An increase in court budget may not have the desired impact on judicial performance [14,15]. Neto et al. [16] show a negative relationship between court expenditure and efficiency scores in Brazilian courts, while results from T. Y. C. Yeung et al. [17] and Melcarne & Ramello [18] indicate the absence of any significant relationship between the two in European countries. Further, results from Voigt & El-Bialy [15] and Fauvrelle & Almeida [19] indicate no relationship between per capita gross domestic product and judicial performance. However, the overall impact of absolute income per capita on efficiency is negative in Polish courts [20]. This may be due to more complex and challenging cases being filed in economically developed regions.

2.2. Information and communication technologies

The usage of information and communication technologies (ICT) can aid courts in multiple ways. Computer facilities can be used in courts to help in administration, management of cases from filing to judgment, provide assistance to judges and facilitate communication between courts and litigants [21]. El Bialy [22] presents a positive impact of computerization on court performance in Egypt, Procopiuck [64] and Fauvrelle & Almeida [19] show a negative impact of electronic filing and processing of cases in Brazil, and Deyneli [21] indicates no significant impact of computerization on the efficiency of European courts. With regards to investments in ICT, some studies such as Gomes et al. [23] and Louro et al. [24] indicate a positive association between expenditures and court productivity, while Sousa & Guimaraes [25] suggests a negative impact. It can be inferred that technological innovations may not always have the expected benefits and should be made after understanding the organization's need.

2.3. Judicial procedures

Common law countries tend to have a lower degree of procedural formalism as compared to civil law countries [26]. Some degree of formalism is needed to ensure accountability of the judiciary [27]. Voigt & El-Bialy [15] use the number of steps that need to be followed for the court to deliver a judgement as a measure of procedural formalism. However, results indicate no relationship between formalism and judicial efficiency. Complicated procedures outlined in the underlying laws or acts can lead to long duration of trials [28]. Procedural ambiguity can also have a similar effect since it would make the procedure or law open to interpretation by judges, which in turn might lead to conflicting judgements by different judges [28]. Judges working on a similar subject matter would need to spend considerable time to study these opposing decisions before deciding the one most appropriate for their case. The composition of judicial demand, namely the proportion of different incoming cases classified according to case matters has a significant impact on judicial efficiency [29]. This implies that procedures that vary according to the case matter influence the supply of justice in terms of court performance.

2.4. Case burden

Courts with a higher burden of cases tend to perform better than courts with a lower burden [10,11,25,30–32]. The judges probably adjust their performance or outputs based on the level of workload. Results from Achenchabe & Akaaboune [33] show that a higher workload enhances efficiency by utilization of economies of scale. Courts with a high caseload per judge also show better performance as compared to courts with low caseload per judge [34,35]. El Bialy [22] and T. Y. C. Yeung et al. [17] suggest contrary results: caseload per judge and backlog per judge have negative effects on efficiency. Bełdowski et al. [20], Espasa & Esteller-Moré [36] and Sousa & Guimaraes [25] show that the output of courts is primarily driven by the demand for judicial services, namely, the number of incoming and pending cases. However, Neto et al. [16] show that courts with low efficiency levels tend to have a higher number of newly filed and pending cases. The number of pending cases at the end of a period has a negative impact on efficiency [33]. The congestion of courts, expressed as the number of cases pending at the beginning of the period also has a detrimental impact on judicial efficiency [36].

2.5. Judges and staff

One would assume that a higher judicial strength, namely, the number of judges would imply a higher judicial output. However, empirical studies indicate a lack of consensus about the actual effect of number of judges on judicial efficiency. Sousa & Guimaraes [25], Rosales-López [11] and Louro et al. [24] show a positive impact of number of judges on the performance of courts. Dimitrova-Grajzl et al. [37] also show a significant positive impact; however, only a limited effect is observed and that also only in a subset of courts. This suggests that judges probably adjust their productivity based on the number of working judges in the court. Bełdowski et al. [20] also show similar results, where number of judges has a significant positive impact specifically for cases that require a full trial. Dimitrova-Grajzl et al. [30] and de Oliveira Gomes et al. [38] indicate the lack of a significant relationship between number of judges and court output. Some studies such as Neto et al. [16], Achenchabe & Akaaboune [33], Christensen & Szmer [39] and Szmer et al. [40] also indicate a negative relationship between judicial performance and number of judges. Beenstock & Haitovsky [34] is one of the pioneer studies revealing the negative impact of number of judges on productivity. Judges under pressure are likely to exert effort, improve their performance and dispose a higher number of cases. If there is a rise in the number of judges in a court, the caseload pressure on the existing judges in the court falls, which leads to a decline in their output. Even though new judges added to the court would contribute to the output, the reduction in the number of cases disposed by existing judges may offset this increase, thereby, reducing overall output.

Various judge characteristics such as age, seniority, education and tenure status may impact judicial efficiency. Ferro et al. [41] indicate that the age of judges has positive relationship with efficiency and their seniority has negative impact. Christensen & Szmer [39] also reveal similar insights and attributes this negative effect to judge burnout. Contrary to this, a higher proportion of experienced judges has a positive effect on efficiency levels in Moroccan courts [33]. El Bialy [22] indicates an insignificant relationship between seniority and efficiency in Egyptian courts. Further, judicial panels with judges from elite law schools, elevated judges from lower courts, visiting senior and appellate judges help attain high efficiency while visiting judges from lower courts have a negative effect on judicial performance [39].

Judges may be transferred on a voluntary or mandatory basis to another office. High judge turnover may lead to reduced efficiency levels on two accounts: first due to flawed management of pending cases by the outgoing judge, and second due to delay in filling the vacated position during which the pending cases would remain suspended. Guerra & Tagliapietra [42], Rosales-López [11] and El Bialy [22] find evidence to support this hypothesis: an increase in judge turnover leads to a decline in judicial performance. Empirical results from Staszkiewicz et al. [43] also indicate that while judges' transfers have a short-term positive impact, they have a long-term negative impact on the judicial system. It is possible that in the short-term, judges want to present good numbers to the stakeholders, hence focus on passing judgements in simple cases that require less time as compared to complex cases demanding more effort.

Many judiciaries provide training to judges at the beginning of their career and offer various courses related to general services, specialized services, computer services during the course of their career. Judicial trainings may be optional or compulsory. Provision of compulsory training to judges is likely to improve judicial performance [15,21]. In line with this, investment in training for judges has a positive impact on efficiency [25]. Even judicial reforms that offer trainings to judges can help improve performance [22].

Judicial staff are court employees that provide direct support to judges. They help foster better working conditions for judges and conduct various administrative tasks. This ultimately allows judges to spend more time on judicial activities, thereby increasing court productivity [38]. Reinforcement of courts in terms of an increase in employees who provide assistance to judges tends to have a positive impact on court output [11,25]. Further, varying impacts of different categories of staff are observed in Italian and Polish courts [20,29].

2.6. Case features

Different features related to the case also affect judicial performance. Cases that are more complex tend to take a longer time to dispose [44]. Cases that are settled have a shorter disposition time as compared to those that go through a full trial [44]. The case category [44], outcomes of cases in terms rulings in favour of the plaintiff or defendant [44], publication of opinions [40] and oral arguments also have a significant relationship with judicial performance [39,40]. Litigants or disputing parties also play a key role in determining the case duration [45]. Intuitively, a higher number of plaintiffs and defendants involved in case may lead to an increase in the duration. However, Bielen et al. [44] find no evidence to support this hypothesis. In case of the litigant type, cases with the government as plaintiff tend to have a shorter duration, while the presence of legal person as a litigant shows no significant relationship with performance.

2.7. Litigious

Litigious refers to the inclination of people or companies to file lawsuits or resorting to the judiciary for minor grievances. Courts located in jurisdictions with high litigious ratio tend to be more efficient [46–48]. The courts have probably modified their management culture or practices to adapt to a higher inflow of cases. Melcarne & Ramello [18] present no impact of litigation rate on court efficiency in a cross-country analysis. However, a negative impact of the public's inclination to sue or commit crimes is observed on court performance in Italy [29].

2.8. Lawyers and legal experts

A higher number of lawyers tends to have a negative effect on judicial efficiency [46]. A greater number of lawyers increases competition and incentivizes them to behave in an opportunistic manner. They are likely to encourage civil litigation even for small claims and induce delay in resolving cases. Mediators and legal experts may be responsible for the delay in resolving cases [44], especially in civil disputes [45].

2.9. Research gap

Every judicial system varies in terms of its organization structure, functioning based on underlying laws and procedural codes, and encompassing environment. Accordingly, the factors influencing judicial performance may also differ across boundaries. Hence, every judiciary requires an independent study to ascertain factors affecting justice delivery. The study aims to develop a broad framework that can be adopted by judicial systems to determine and rank factors relevant to their context. Majority literature on Indian courts focusses on the effect of judicial efficiency on the country's growth [28,49]. Limited work has been done to evaluate the impact of different factors on the performance of Indian judiciary [28,32,35]. Further, as observed from literature, the influence of factors in terms of direction and degree is not uniform across judicial systems; the impact of same factors may be positive, negative or none at all. This study aims to identify different factors that affect the functioning of Indian courts and determine their degree of influence. An analysis of various factors and their relative impact helps us identify specific areas that have scope of improvement so that appropriate policy interventions can be informed for an efficient judicial system. We apply fuzzy best-worst method, a multi-criteria decision-making tool in this analysis. To the best of our knowledge, this is the first ever application of fuzzy best-worst method in the judicial domain. Section 3 presents the methodology adopted in the study.

3. Methodology

An overview of the methodological framework to determine factors affecting judicial efficiency is presented in Fig. 1. The framework is divided into three main steps: identification of possible factors, categorization of finalized factors, and prioritization of factors. The respondents in the methodology should be experts representing various facets of the judicial system to ensure that different views and insights are included in the study. Experts should be consulted during each step of the study. The first step consists of a review of literature to create a list of possible factors relevant to the context. These factors should be informally discussed with respondents. The list of factors can be finalized based on the experience and insights of the experts. In the second

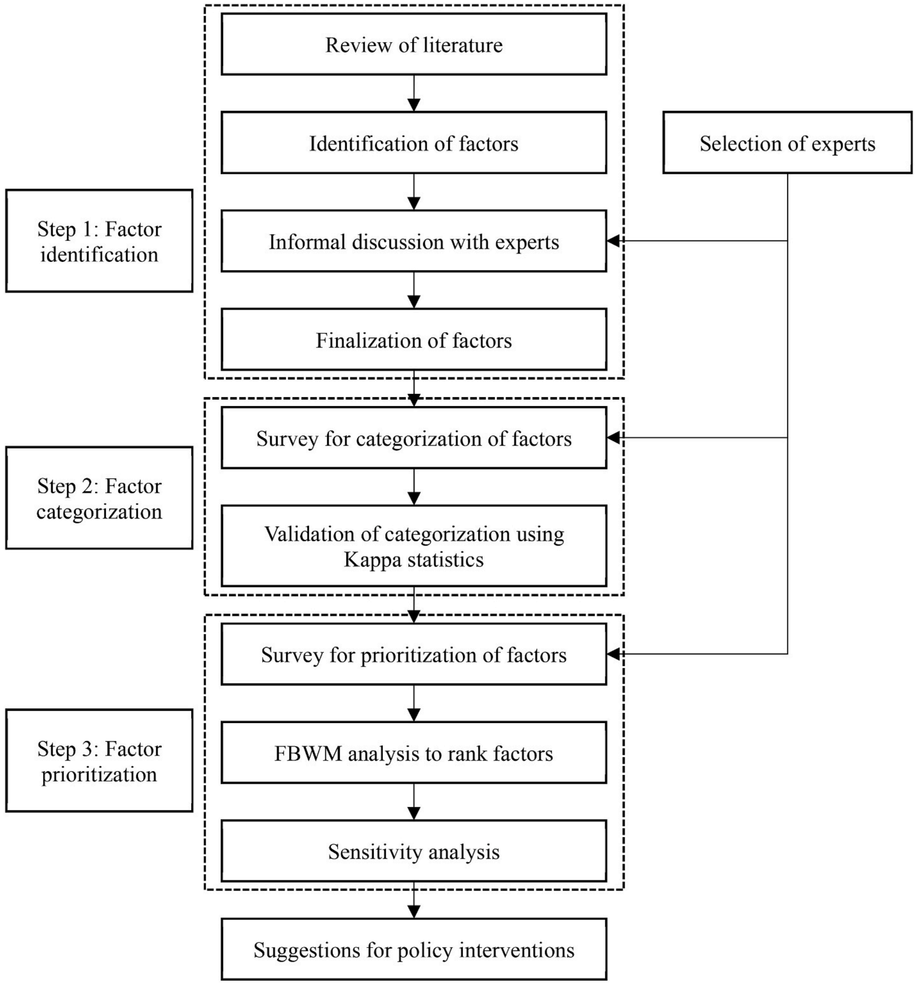

Fig. 1. Research methodology.

step, the factors should be grouped into broad categories. This can be done through a questionnaire circulated to the respondents followed by a validation of the categorization using Kappa statistic. The third step consists of a questionnaire administered to rank the categories and factors within each category using fuzzy best-worst method. This helps determine an overall ranking of the factors. A sensitivity analysis can also be conducted to test the robustness of the model. Policy interventions can be suggested to improve judicial performance based on the ranking of the factors. The methodology is applied to the Indian judiciary. The panel of experts chosen for the study are described in section 3.1. Section 3.2 discusses the method adopted for categorization of factors, namely, Kappa statistic. Section 3.3 describes the methodology adopted for ranking the factors, namely, FBWM.

Table 1
Profile of experts.

Expert no.	Respondent domain	Years of experience
1	Lawyer (Practising in district court and high court)	8
2	Lawyer (Corporate lawyer, practicing in NCLT)	2
3	Lawyer (Corporate lawyer, practising in NLCT and high court)	12
4	Researcher (Judicial system)	3
5	Researcher (Judicial system)	2
6	Researcher (Law and judicial system), lawyer	15
7	Researcher (Law and judicial system), lawyer	11
8	Judge	20
9	Public policy professional	6
10	Professor (Law university), lawyer	7

3.1. Selection of experts

The methodology should include the opinion of various stakeholders in the judicial domain such as lawyers, judges, law professors and researchers. It should incorporate the opinions of both young as well as experienced professionals working in the judicial space. The profile of experts surveyed for the case of India is provided in Table 1. We have included lawyers working at different courts and tribunals to enable us to capture issues plaguing different levels of the judicial system. We have added a lawyer practicing in both district and high courts, a lawyer primarily working in the corporate sector and practicing in the National Company Law Tribunal (NCLT), and a corporate lawyer practicing in the NCLT and high court. While lawyers are actively involved in the functioning of courts, judicial research has become relevant in the current scenario of high pendency levels. To this end, we have added two researchers working primarily on the Indian judicial system in our panel. We have also included two lawyer-cum-researchers engaged in research on Indian laws and judiciary. In addition to experience gained from practicing in courts, these professionals helped provide key insights from their research. This aided in the reduction of bias, since they are not directly involved in court proceedings. We have included a working judge with over two decades of experience in our list of experts, since judges form an integral part of delivering justice. Further, we have also added a public policy professional to help understand the policy perspectives about judicial functioning. Lastly, we enlisted a professor from a premier law institute of the country in our panel since it is the teachers who assist in shaping the minds of the next generation of judicial officials.

The discussions and interviews with experts were conducted in both online and offline modes depending on their availability. All respondents were informed at the beginning of the surveys that the terms 'judicial efficiency' and 'judicial performance' in the context of this study represent the quantitative performance of courts. Both the questionnaires were created on Google forms and the corresponding links were shared with the experts. The questionnaires used in the study are provided in supplementary material. It took about 15 min to administer the first questionnaire (categorization) and around 45 min to conduct the second interview (prioritization). The surveys were conducted during March 2023.

3.2. Identification and categorization of factors

The initial list of factors was created by conducting a thorough review of literature followed by a shortlisting of factors relevant to the Indian judicial system. This list was then discussed with respondents through semi-structured interactions. These discussions led to final list of twenty factors. The factors are described in detail in section 4. It is often difficult to apply an MCDM technique when we are working with numerous factors. In such scenarios, it is preferred that similar factors are grouped together into few broad categories. Once the grouping is done, an appropriate statistical test is applied to validate the categorization. We identified five broad categories appropriate for our study through a literature review followed by discussions with experts. The categories are described in section 4.1. The experts were surveyed to group the twenty factors into these five categories. Their responses were validated using Kappa statistic.

Kappa statistic is a commonly used measure to test the consensus between decision-makers across different domains [50–53]. It was developed by Cohen [54] and is based on the concept of interrater reliability [55]. The Kappa statistic measures the percent agreement by considering that some agreements may occur purely by chance. Mathematically, it is calculated as the ratio between difference of observed theoretical agreement and chance agreement to the possible agreement beyond chance [52].

The formulae for the calculation of Kappa statistic are as follows:

$$F_i = \frac{1}{N(N-1)} \sum_{j=1}^{R} \left(e_{ij}^2 - e_{ij}\right) \tag{1}$$

$$C_j = \frac{\sum_{i=1}^{Q} e_{ij}}{\sum_{j=1}^{R} \left(\sum_{i=1}^{Q} e_{ij} \right)} \tag{2}$$

$$T = \frac{1}{Q} \sum_{i=1}^{Q} F_i \tag{3}$$

$$P = \sum_{j=1}^{R} C_j^2 \tag{4}$$

$$\kappa = \frac{(T-P)}{(1-P)} \tag{5}$$

where N is the number of experts or decision-makers, Q is the number of factors to be categorized, R is the number of broad categories, e_{ij} is the evaluation of factor i with respect to category j, F_i is the percentage involvement of factor i, C_j is the percentage alignment to category j, T is the probability of observed theoretical agreement, P is the probability of chance agreement, and κ is the Kappa statistic.

The range of Kappa statistic (equation (5)) is from −1 to 1, which indicates no consensus to perfect consensus. The level of consensus and corresponding ranges of the index are as follows: no consensus for values between −1 and 0, slight consensus from 0.01 to 0.20, moderate consensus from 0.21 to 0.40, reasonable consensus from 0.41 to 0.60, substantial consensus from 0.61 to 0.80, and perfect consensus from 0.81 to 1.

3.3. Ranking of factors

Best-worst method (BWM) is a method developed by Rezaei [57] to solve MCDM problems. In this method, the decision-maker chooses the best and worst criteria according to their judgement. They are then asked to conduct pairwise comparisons of other criteria with the best and worst criteria to determine the weights of various criteria. BWM offers the following advantages over other MCDM techniques: i) less comparison data is required, and ii) the comparisons are more consistent in nature. Further, a sample size of five to seven is adequate for consistent results [57,58]. However, human subjectivity associated with the process and lack of complete information can make it difficult to model the problem with just crisp values of the criteria. To account for this ambiguity and vagueness, Guo & Zhao [58] developed fuzzy best-worst method (FBWM). FBWM is an expansion of the traditional BWM to a fuzzy environment. The crisp preferences of decision-makers are expanded to triangular fuzzy numbers by using linguistic terms for pairwise comparisons. FBWM is a widely used tool across different sectors such as supply chains [52], power distribution [59], energy [60] and transportation [51]. In fact, FBWM shows higher comparison consistency as compared to BWM [58].

The detailed steps for FBWM are as follows [58]:

1. Establish set of decision criteria: A set of decision criteria $\{c_1, c_2, \ldots, c_n\}$ is determined for the decision-making process.
2. Determine the best and worst criteria: In this step, the best (most important) and worst (least important) criteria are identified by the decision-maker. The best and worst criteria are represented by c_{B} and c_{W}, respectively.
3. Conduct fuzzy reference comparisons for the best criterion: In this step, fuzzy reference comparisons of the best criterion are determined with respect to the other criteria. $\widetilde{\mathrm{A}}_{\mathrm{B}} = (\widetilde{\mathrm{a}}_{\mathrm{B}1}, \widetilde{\mathrm{a}}_{\mathrm{B}2}, \ldots, \widetilde{\mathrm{a}}_{\mathrm{B}n})$ represents the best-to-others vector, where $\widetilde{\mathrm{a}}_{\mathrm{B}j}$ represents the fuzzy preference of the best criterion c_{B} over criteria j, $(j = 1, 2, \ldots n)$. The fuzzy preferences in linguistic terms are converted to triangular fuzzy numbers according to rules presented in Table 2 [61] and $\widetilde{\mathrm{a}}_{\mathrm{BB}} = (1, 1, 1)$.
4. Conduct fuzzy reference comparisons for the worst criterion: In this step, fuzzy reference comparisons of all criteria are determined with respect to the worst criterion. $\widetilde{\mathrm{A}}_{\mathrm{w}} = (\widetilde{\mathrm{a}}_{1\mathrm{w}}, \widetilde{\mathrm{a}}_{2\mathrm{w}}, \ldots, \widetilde{\mathrm{a}}_{n\mathrm{w}})$ represents the others-to-worst vector, where $\widetilde{\mathrm{a}}_{j\mathrm{w}}$ represents the fuzzy preference of the criteria j, $(j = 1, 2, \ldots n)$ over the worst criterion c_{W}. The fuzzy preferences in linguistic terms are converted to triangular fuzzy numbers according to rules presented in Table 2 and $\widetilde{\mathrm{a}}_{\mathrm{ww}} = (1, 1, 1)$.
5. Determine the optimal weights of the criteria $(\widetilde{\mathrm{w}}_1^*, \widetilde{\mathrm{w}}_2^*, \ldots, \widetilde{\mathrm{w}}_n^*)$: For fuzzy preferences, say $\widetilde{\mathrm{a}}_{\mathrm{B}j}$ and $\widetilde{\mathrm{a}}_{j\mathrm{w}}$, the objective is to determine weights that minimize the absolute maximum gap

Table 2
Consistency indices for FBWM.

Linguistic terms	Triangular fuzzy number	Consistency index
Just equal	(1,1,1)	3.00
Equally influential	(1/2,1,3/2)	3.80
Slightly influential	(1,3/2,2)	4.56
Moderately influential	(3/2,2,5/2)	5.29
Very influential	(2,5/2,3)	6.00
Extremely influential	(5/2,3,7/2)	6.69

$|\widetilde{w}_B/\widetilde{w}_j - \widetilde{a}_{Bj}| and |\widetilde{w}_j/\widetilde{w}_W - \widetilde{a}_{jW}|$ for all j (equation (6)). The sum of weights is equal to unity as depicted in equation (7), the fuzzy weight of criteria j is represented by $\widetilde{w}_j = (l_j^w m_j^w u_j^w)$ in equation (8), and non-negative constraints in equation (9). The FBWM model can be solved to determine optimal weights.

$$\underset{j}{Min\,Max}\left\{\left|\widetilde{w}_B \big/ \widetilde{w}_j - \widetilde{a}_{Bj}\right|, \left|\widetilde{w}_j \big/ \widetilde{w}_W - \widetilde{a}_{jW}\right|\right\} \tag{6}$$

Subject to:

$$\sum_{j=1}^{n} R(\widetilde{w}_j) = 1 \tag{7}$$

$$l_j^w \leq m_j^w \leq u_j^w \forall j \tag{8}$$

$$l_j^w \geq 0 \forall j \tag{9}$$

The model can be rewritten by modifying the objective function (equation (6)) to equation (10) and adding constraints (equations (11) and (12)) to the existing constraints (equations (7) to (9)).

$$Min \widetilde{\xi} \tag{10}$$

Subject to:

$$\left|\widetilde{w}_B \big/ \widetilde{w}_j - \widetilde{a}_{Bj}\right| \leq \widetilde{\xi} \forall j \tag{11}$$

$$\left|\widetilde{w}_j \big/ \widetilde{w}_W - \widetilde{a}_{jW}\right| \leq \widetilde{\xi} \forall j \tag{12}$$

where $\widetilde{\xi} = (l^{\xi}, m^{\xi}, u^{\xi})$. Considering $(l^{\xi} \leq m^{\xi} \leq u^{\xi})$ and assuming $\widetilde{\xi}^* = (k^*, k^*, k^*)$, $k^* \leq l^{\xi}$ the modified model to determine optimal weights is as follows:

$$Min \widetilde{\xi}^* \tag{13}$$

Subject to:

$$\left|\frac{l_B^w m_B^w u_B^w}{l_j^w m_j^w u_j^w} - \left(l_{Bj}, m_{Bj}, u_{Bj}\right)\right| \leq (k^*, k^*, k^*) \tag{14}$$

$$\left|\frac{l_j^w m_j^w u_j^w}{l_W^w m_W^w u_W^w} - \left(l_{jW}, m_{jW}, u_{jW}\right)\right| \leq (k^*, k^*, k^*) \tag{15}$$

$$\sum_{j=1}^{n} R(\widetilde{w}_j) = 1 \tag{16}$$

$$l_j^w \leq m_j^w \leq u_j^w \forall j \tag{17}$$

$$l_j^w \geq 0 \forall j \tag{18}$$

The optimal fuzzy weights can then be transformed to crisp weights using graded mean integration representation (equation (19)).

$$R(\widetilde{a}_i) = \frac{l_i + 4m_i + u_i}{6} \tag{19}$$

Consistency ratio is used to test the degree of consistency of fuzzy pairwise comparisons. A comparison is said to be fully consistent if $\widetilde{a}_{Bj} * \widetilde{a}_{jW} = \widetilde{a}_{BW}$, where $\widetilde{a}_{BW}$ is the fuzzy preference of best criterion over the worst, $\widetilde{a}_{Bj}$ is the fuzzy preference of the best criterion over criterion j, and $\widetilde{a}_{jW}$ is the fuzzy preference of criterion j over the worst criterion. Inconsistency will be highest when $\widetilde{a}_{Bj} = \widetilde{a}_{jW} = \widetilde{a}_{BW}$. Mathematically, the consistency ratio is the maximum possible value of ξ when equation (20) is evaluated for all values of u_{BW}. Here, u_{BW} is the upper bound of $\widetilde{a}_{BW} = (l_{BW}, m_{BW}, u_{BW})$ and ξ is a crisp value of $\widetilde{\xi}$. One can refer to Guo & Zhao [58] for further details.

$$\xi^2 - (1 + 2u_{BW})\xi + \left(u_{BW}^2 - u_{BW}\right) = 0 \tag{20}$$

where $u_{BW} = 1, 3/2, 5/2, 7/2$ and $9/2$, respectively.

4. Results

The framework depicted in Fig. 1 is applied to the Indian judicial context. We identified numerous factors that may impact judicial performance through a review of literature. Informal discussions were conducted with the experts. We finalized a list of 20 factors based on the inputs and insights of the experts. Table 3 enumerates the factors along with their description and references.

The factors are supported by literature and adjusted according to the Indian context after discussions with experts. For example, Indian courts are known for their judicial burden and slow case resolution. Hence, experts suggest that a high backlog for judges will lead to low judicial efficiency. Similarly, it is often cited that low working strength of judges is the primary reason behind poor judicial output. Accordingly, we hypothesize that a low number of judges would be a barrier to enhanced judicial performance. Few experts said that judges have insufficient knowledge about subject matters of cases listed on their docket, leading to delay in resolution of cases. This may be due to lack of experience and training of judges. Cases wherein a government body is a litigant form a significant proportion of pending cases in Indian courts [62,63]. Hence, they are likely to have a negative relationship with judicial performance. Litigation rate and crime rate represent judicial demand in terms of inflow of cases, and hence, we hypothesize that high rates will lead to slow judicial functioning. In terms of income per capita, we expect that as the purchasing power increases, more people are inclined to resort to courts for resolution of disputes, thereby reducing efficiency. Literacy rate represents a socio-demographic factor. We assume that low literacy rates lead to insufficient understanding of the legal system by litigants, which ultimately causes prolonged case durations. We added inadequate physical infrastructure as a factor based on our discussions with experts. Multiple experts said that a lack of court rooms for judges and other physical infrastructure, especially in lower courts, is a major issue across the country.

The classification of the above-mentioned factors is presented in section 4.1. The weights and ranking of factors from the FBWM survey are presented in section 4.2, followed by a sensitivity analysis of the results in section 4.3.

4.1. Categorization of factors

As described in section 3.2., the application of an MCDM technique with many factors can become complex. Hence, we aim to classify the twenty factors into few broad categories. We determine five categories through a review of literature and discussions with respondents. The categories are as follows:

- Organizational factors include factors related to organizational issues in the judiciary and are within the court's ambit
- Judge-related factors comprise of factors related to judges, their position and career
- Case-related factors consist of factors pertaining to features or details of cases
- Procedural factors include factors related to the procedures followed in courts to resolve cases
- Exogenous factors consist of factors that are outside the control of the judiciary

The experts were surveyed to segregate the factors into different categories based on their knowledge and experience. The classification responses were tested for the strength of agreement between the experts using Kappa statistic. The value of Kappa statistic obtained was 0.62 which indicates substantial consensus among the respondents. The responses and calculations related to Kappa statistic are provided in

Table 3
List of factors.

Factor	Description	References
Unfair distribution of funds	Inequitable distribution of financial resources	Peyrache & Zago [13], Viapiana [12]
Lack of IT initiatives and infrastructure	Scarce adoption of information technology (IT) practices and inadequate IT infrastructure	Deyneli [21], El Bialy [22], Fauvrelle & Almeida [19], Procopiuck [64]
Low staff per judge	Few court employees providing direct support to judges	Bełdowski et al. [20], de Oliveira Gomes et al. (2016), G. Falavigna & Ippoliti [29], Rosales-López [11], Sousa & Guimaraes [25]
Inadequate physical infrastructure	Lack of physical infrastructure such as court rooms	Discussion with experts
High backlog for judges	High number of pending and incoming cases for judges	Achenchabe & Akaaboune [33], Beenstock & Haitovsky [34], Bełdowski et al. [20], Dimitrova-Grajzl et al. [30], El Bialy [22], Espasa & Esteller-Moré [36], Grajzl & Silwal [10], Grajzl & Silwal [31], Gupta & Bolia [35], Neto et al. [16], Rosales-López [11], Sen [32], Sousa & Guimaraes [25], T. Y. C. Yeung et al. [17]
Inadequate judicial training	Lack of sufficient training provided to judges	Deyneli [21], El Bialy [22], Sousa & Guimaraes [25], Voigt & El-Bialy [15]
Less number of judges	Inadequate number of working judges in courts	Achenchabe & Akaaboune [33], Beenstock & Haitovsky [34], Bełdowski et al. [20], Christensen & Szmer [39], de Oliveira Gomes et al. (2016), Dimitrova-Grajzl et al. [37], Dimitrova-Grajzl et al. [30], Louro et al. [24], Neto et al. [16], Rosales-López [11], Sousa & Guimaraes [25], Szmer et al. [40]
Frequent judicial transfers	Regular transfers of judges amongst courts	El Bialy [22], Guerra & Tagliapietra [42], Rosales-López [11], Staszkiewicz et al. [43]
Judges' experience	Lack of experience of judges	Achenchabe & Akaaboune [33], Christensen & Szmer [39], El Bialy [22], Ferro et al. [41]
High case complexity	High complexity associated with cases	Bielen et al. [44]
High number of litigants	High number of plaintiffs and defendants in cases	Bielen & Marneffe [45], Bielen et al. [44]
Government as a litigant	Government as a litigant in cases	Bielen et al. [44], Kinhal & Kumar [63], Kinhal et al. [62]
Procedural complexities	Complicated procedures outlined in the underlying acts and codes	Chemin [28], G. Falavigna & Ippoliti [29], Voigt & El-Bialy [15]
Procedural ambiguities	Flexibility in interpretations of rules and procedures	Chemin [28]
Mediation activities	Court-mandated mediation activities	Bielen et al. [44]
Litigation rate	Number of cases filed per population	Castro & Guccio [46], Giacalone et al. [47], G. Falavigna & Ippoliti [29], Melcarne & Ramello [18], Nissi et al. [48]
Crime rate	Number of crimes reported per population	El Bialy [22], G. Falavigna & Ippoliti [29]
Income per capita	Measure of the amount of money earned per person	Bełdowski et al. [20], Fauvrelle & Almeida [19], Voigt & El-Bialy [15]
Literacy rate	Proportion of people aged 7 and above who are literate	Achenchabe & Akaaboune [33]
High number of lawyers per population	High number of registered lawyers per population	Castro & Guccio [46]

Table 4, followed by the final classification of factors in Table 5.

4.2. Ranking of factors

Once the factors were grouped into different categories, FBWM was used to determine the weights of factors and categories. A survey was administered in which each expert was asked to identify the most important and least important categories that affect judicial efficiency. As per the FBWM approach outlined in section 3.3., these represent the best and worst criteria, respectively. After this, the experts were asked to do a pairwise comparison of each category with the most and least important categories in terms of their relative impact. The same procedure was repeated for factors within each category. The pairwise comparisons were modelled as non-linear programming models and solved using LINGO software (version 20.0). The models determined optimal fuzzy weights, which were converted to crisp weights. Geometric means were used to determine aggregate local weights for each category as well as factor. The global weight of each factor was calculated as the product of the local weight of the factor and local weight of the category to which the factor belongs. The ranks, global weights and local weights of the factors and categories are presented in Table 6. The consistency ratio was also calculated for all pairwise comparisons. All values of consistency ratios are close to 0, indicating that our results are consistent. The insights from these results are discussed in section 5.

4.3. Sensitivity analysis

We conduct a sensitivity analysis to test the robustness of the model. The process adopted is similar to the sensitivity analysis performed by Bhatnagar et al. [51] and Patil, Shardeo, Dwivedi & Madaan [56], wherein the local weights of categories and factors are varied. We consider ten scenarios (SC1 to SC10) for each sensitivity analysis. The first scenario, SC1 corresponds to the weights determined by our analysis. In scenarios SC2 to SC10, the local weight of the highest ranked category is varied from 0.1 to 0.9, and the weights of the other categories are modified accordingly. The same process is repeated for factors under each category. Fig. 2 (a) - (f) presents the ranks of categories and factors for all scenarios. Consider the case of judge-related factors (Fig. 2 (c)). The most important factor as per our model is high backlog for judges. Its rank remains the same in seven out of the nine scenarios in which the local weights are varied. In fact, the ranks of other factors also remain consistent in at least seven of the nine scenarios. Overall, we observe that the ranks of most categories and factors within categories remain constant. Hence, we conclude that our model is robust.

5. Discussion and policy implications

Judicial systems across countries differ in terms of judicial independence, political control, legal culture, judge caseload, underlying laws and procedures. The factors that affect the performance of courts vary across boundaries and so do their direction of influence. Hence, judicial reforms cannot be simply extrapolated from one country to another; interventions that work for a given judicial system may not work for another. The interventions should be developed according to various factors affecting judicial efficiency and their relative impact. We develop a framework to identify these factors and apply it to the case of Indian courts.

The results from our study indicate that amongst the broad categories, organizational factors are the most important followed by

Table 4
Responses from survey for categorization.

Factor	Organizational factors	Judge-related factors	Case-related factors	Procedural factors	Exogenous factors
Unfair distribution of funds	8	0	0	0	2
Low staff per judge	6	4	0	0	0
Inadequate physical infrastructure	9	0	0	0	1
High number of litigants	0	0	8	1	1
Litigious culture	0	0	1	0	9
Inadequate judicial training	2	7	0	0	1
Government as a litigant	0	0	8	1	1
Lack of IT initiatives and infrastructure	10	0	0	0	0
Less number of judges	4	6	0	0	0
Procedural complexities	0	0	1	9	0
Income per capita	0	0	0	0	10
Judge's experience	2	8	0	0	0
High number of lawyers per population	1	0	0	1	8
Crime rate	1	0	0	0	9
High backlog for judges	2	8	0	0	0
Frequent judicial transfers	4	6	0	0	0
High case complexity	0	0	8	2	0
Mediation activities	0	0	2	8	0
Literacy rate	0	0	0	0	10
Procedural ambiguities	0	0	1	9	0
C_j (Percentage alignment)	0.245	0.195	0.145	0.155	0.260

Table 5
Final categorization of factors.

Category	Factor	Code
Organizational	Unfair distribution of funds	O1
	Lack of IT initiatives and infrastructure	O2
	Low staff per judge	O3
	Inadequate physical infrastructure	O4
Judge-related	High backlog for judges	J1
	Inadequate judicial training	J2
	Less number of judges	J3
	Frequent judicial transfers	J4
	Judges' experience	J5
Case-related	High case complexity	C1
	High number of litigants	C2
	Government as a litigant	C3
Procedural	Procedural complexities	P1
	Procedural ambiguities	P2
	Mediation activities	P3
Exogenous	Litigation rate	E1
	Crime rate	E2
	Income per capita	E3
	Literacy rate	E4
	High number of lawyers per population	E5

procedural and case-related factors (Table 6). The ten most important factors also belong to these three groups. Within the categories, procedural complexities (P1), lack of IT initiatives and infrastructure (O2), high case complexity (C1), low staff per judge (O3) and procedural ambiguities (P2) are ranked the highest. Hence, improvements in these factors are likely to yield significant enhancements in judicial efficiency. We suggest policy interventions for each category in sections 5.1. to 5.5. Section 5.6. concludes the section with a discussion on the analysis of interventions.

5.1. Organizational factors

Information and communication technologies (ICT) are essential to the working of judicial systems, where there are multiple stakeholders, each with their own demands and expectations from e-justice services. The application of ICT in judicial systems can lead to better coordination, faster retrieval of information and statistics, reduction in case delay, automated case distribution amongst judges, fair workload, improved productivity, faster registration of cases, information security, efficient and effective judicial performance, and overall trust and transparency [65–73].

Table 6
Weights and ranks of factors.

Category	Local weight	Factor	Local weight	Global weight	Rank
Organizational	0.2672	Unfair distribution of funds (O1)	0.2164	0.0578	7
		Lack of IT initiatives and infrastructure (O2)	0.3132	0.0837	2
		Low staff per judge (O3)	0.2623	0.0701	4
		Inadequate physical infrastructure (O4)	0.2082	0.0556	8
Judge-related	0.1859	High backlog for judges (J1)	0.2513	0.0467	11
		Inadequate judicial training (J2)	0.1765	0.0328	13
		Less number of judges (J3)	0.2479	0.0461	12
		Frequent judicial transfers (J4)	0.1538	0.0286	16
		Judges' experience (J5)	0.1704	0.0317	15
Case-related	0.1888	High case complexity (C1)	0.3899	0.0736	3
		High number of litigants (C2)	0.3332	0.0629	6
		Government as a litigant (C3)	0.2768	0.0523	10
Procedural	0.2230	Procedural complexities (P1)	0.4685	0.1045	1
		Procedural ambiguities (P2)	0.2840	0.0633	5
		Mediation activities (P3)	0.2475	0.0552	9
Exogenous	0.1351	Litigation rate (E1)	0.2092	0.0283	17
		Crime rate (E2)	0.1999	0.0270	19
		Income per capita (E3)	0.2015	0.0272	18
		Literacy rate (E4)	0.2407	0.0325	14
		High number of lawyers per population (E5)	0.1488	0.0201	20

The Government of India launched the "eCourts Integrated Mission Mode Project" in 2007 for ICT development of the Indian judiciary. Phases I and II of the project were implemented in 2011 and 2015, respectively. Various technological initiatives have been introduced under the project [74]. These include, but are not limited to:

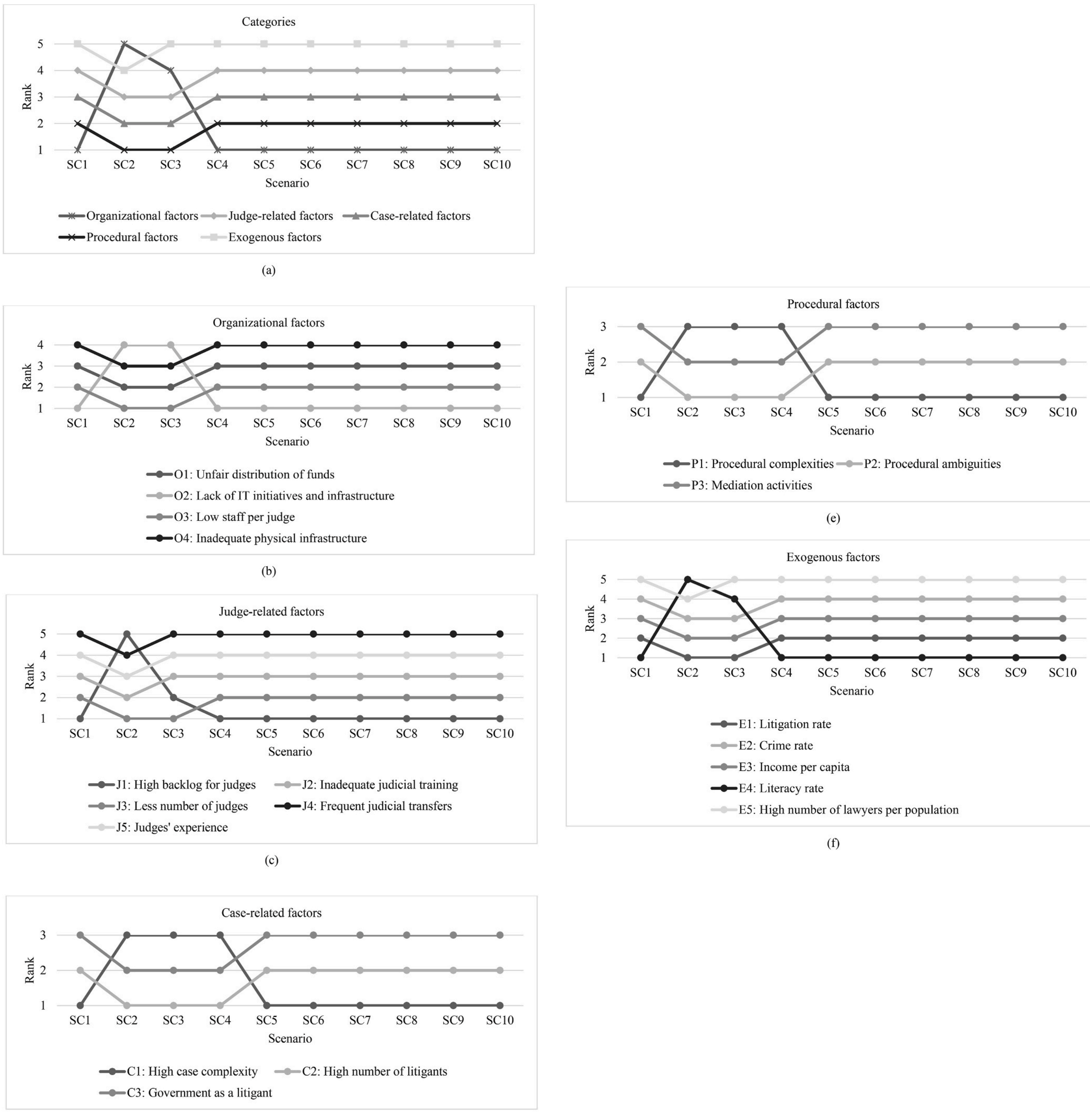

Fig. 2. (a)–(f): Sensitivity analysis.

computerization of lower courts (physical infrastructure and internet connectivity), national databases of case information, status and orders such as e-courts portal (ecourts.gov.in) and National Judicial Data Grid (https://njdg.ecourts.gov.in/), independent websites by lower courts, Case Information Software for courts, virtual courts, portals to provide real-time information about case status, cause list, and judgements to lawyers and litigants through messages, email, and information kiosks, mobile apps for lawyers and judges, video conferencing facilities in courts, e-filing of cases, manuals and videos for guidance about e-filing, trainings and awareness programs for court officials. The objective of the latest phase of the project, namely Phase III is to make the justice system more accessible, efficient and transparent for people seeking judicial remedies as well people who are part of the system [75].

The Indian judiciary has made significant advancements through the e-courts project. However, our study shows that lack of IT initiatives and infrastructure is presently a major factor inhibiting judicial efficiency. It is possible that the reforms have not been realized to their full potential and there is still a lot to achieve. The vision document of the third phase highlights various challenges remaining after the implementation of ICT reforms in the first two phases. Some challenges and possible measures to address the same are discussed subsequently.

There is a network connectivity divide across the country. Some states have good internet connectivity, while some lack even the basic infrastructure. Technical issues due to lack of supporting infrastructure can be major deterrents in the successful implementation of IT reforms [66,72,73,76,77]. Fast internet connectivity and uninterrupted power supply need to be ensured for seamless storage and exchange of large volumes of data [78].

The training provided to court staff about different modules of e-courts is considered inadequate. Litigants and lawyers are also unaware of many facilities offered by e-courts. Judges and staff need to be provided training courses for better utilization of IT systems [77,78]. The public can be encouraged to use e-justice platforms by making them aware of its benefits and ease of use [78]. Data handling in terms of security, confidentiality and integrity is an important concern in a judicial system due to the sensitive nature of the information [77,78]. Phase III of the project also aims to balance security and privacy along with maintaining court openness. Some information can be restricted from public access so that only concerned parties have access to this information [79].

A feedback mechanism from various stakeholders is missing which leads to adoption issues by users. There are problems associated with process re-engineering and integration of technology. There is also a lack of uniformity across courts. For example, high courts across the country have different processes and websites. Hence, the IT systems should be properly designed and developed. A system designed without taking the inputs from the end users will lead to heavy costs without any significant benefits [78]. Committees consisting of different stakeholders can be set up to convey judicial requirements and concepts to IT experts [79,80]. Evaluation models can also be developed to assess the performance of e-justice from a user's perspective [66,81,82]. The information systems of the judiciary can be integrated with other e-government portals and agencies such as banks, revenue systems and transport departments [68,78]. Legislative changes should be made prior to introduction of reforms to address legal concerns [70,78]. Lastly, it is important to note that ICT reforms may take time to align with policies and continuous improvements may be needed to realize their full potential.

The judicial system can use performance-based budgeting models wherein performance indicators or outputs are used to determine the budget or funds allocated to a public system. The models are capable of improving allocative and technical efficiencies of courts [83]. However, they should be used with caution: judiciaries cannot just focus on quantitative outputs, they also need to ensure fairness of procedures and quality decisions. Judges may feel pressurised to complete the targets, even at the expense of delivering quality judgements. Hence, both qualitative and quantitative aspects should be considered while establishing performance-based budgets. Further, the judiciary can be asked for a budget refund in case the expected targets are not met [83]. Judicial autonomy can be enhanced in terms of finances and operations through the establishment of a separate council for the judiciary [84].

The infrastructural capacities of courts need to be enhanced. Once there are sufficient court rooms for judges to preside over, corresponding increases in other resources such as staff and IT systems can have a meaningful impact. To combat low staff levels, staff requirements according to courts and judges should be first estimated. Depending on the demand, staff may be recruited at specific levels [29], or in certain courts [85] for enhanced judicial efficiency.

5.2. Procedural factors

Procedural complexities and ambiguities play a significant role in the time taken to process cases through courts, thereby affecting judicial efficiency. Policy reforms should be aimed at simplifying and shortening court procedures [67,85–87]. This will ultimately help enhance court speed and reduce case backlog. For example, empirical research supports the positive impact of a civil procedures reform on reducing the disposition time of cases in Belgian courts [44]. Improvements in civil and criminal procedures, namely the practices adopted by judges and staff can also help improve judicial efficiency [29,88]. A separate set of procedures can be established for large and complex cases [87]. Time limits can be set for judicial processes [87]. The overall procedure of a case can be divided into multiple steps to ascertain the steps that have the most influence on efficiency [89]. The independent impact of the time spent at each step can be measured so that policy makers know which step has the potential for maximum improvement.

Procedural laws also play a crucial role in the operation and impact of substantive laws [90]. For example, if a borrower defaults on a loan, substantive laws protect the lender with property rights and empower them to recover the loss. However, it is the procedural laws that dictate the actual process to recover the bad loan. In this case, strong procedural laws can facilitate a quick recovery process, thereby reducing delay and associated costs. Hence, established procedures should be supportive of the substantive laws to ensure that rights are not just protected, but they are enforced in a timely manner.

5.3. Case-related factors

Specific case types should be prioritised and studied in detail to identify the reasons for their high inflow, pendency levels or long durations. Reforms with specific aims can then be developed and implemented. For case types wherein the processing time is generally high, the underlying laws and procedures can be studied to determine bottlenecks in the process. A separate set of procedures can be established for large and complex cases [87]. For cases that have a high inflow, the court can reduce the admission rates through limitations [2,91]. Certain case types can also be discontinued or disallowed in courts to reduce pending cases [91]. However, this might not be feasible solution requiring significant changes in the constitution.

Litigants or disputing parties play a key role in determining the case duration wherein they spend the maximum time in preparing their pleadings followed by postponement of hearings [45]. This effect is likely to multiply manifold when there are high number of litigants. However, the judge is capable of reducing the duration of these activities by adopting a more active role.

Caseflow management practices can be established to enhance judicial output. Caseflow management refers to the combination of resources, techniques and procedures essential to an efficient handling of a case from its filing to resolution [1]. Judges and the administrative staff in courts play a crucial role in ensuring fruitful caseflow management. A few key aspects of caseflow management are as follows [1,87]:

- Managing case progress: the courts need to pay attention to the movement of a case through various events, ensuring that critical events are prioritised and given timely hearings. Courts can have a case scheduling system so that the schedule of case events or hearings is given to the involved parties beforehand, thereby ensuring that the case is processed in a timely manner.
- Meaningful court events: the courts need to communicate to all concerned parties that all case events have a purpose and deadlines need to be met ensuring timeliness in the resolution of cases.
- Avoiding unnecessary continuances: the courts should ensure that only a reasonable number of continuances are sought during the course of a case and frivolous continuances are avoided. The courts can also limit the number of continuances to accommodate genuine issues faced by the litigants.
- Differentiated case management: different types of cases require different processes and should be prioritised accordingly.
- Collaborative approaches: different agencies such as police, courts and prosecution need to adopt more collaborative practices to ensure better information exchange and case management.

5.4. Judge-related factors

Results indicate that judge-related factors are considered less important. Surprisingly, even the factor related to number of judges is ranked lower. As indicated in literature, adding more judges may not always have the expected impact on performance. Some studies support the view that an increase in judge strength will lead to a reduced caseload, improved speed and quality of decisions [39,92,93]. In fact, in many judiciaries, the sanctioned strength of judges is sufficient to meet judicial demand; it is the vacant judicial positions that need to be filled to reduce judicial burden [2]. This may be the case for Indian courts as well and reducing the time taken to fill judicial vacancies might be a good solution [39]. However, more judicial positions can lead to increased fixed costs in terms of salary and benefits to judges, their staff, infrastructural requirements and associated expenditures such as building and maintenance of courtrooms [94]. Hence, caution should be exercised while adding judges. The possible impact of adding judges on judicial performance can be estimated based on previous data [91]. Accordingly, the approximate number of additional judges needed to achieve desired performance levels can be determined.

Results also indicate that judicial transfers are not detrimental to judicial efficiency. Transfer of judges among courts can have both quantitative and qualitative benefits for the judiciary. Quantitatively, it can help meet judicial demand or capacity needs while avoiding over and under-utilization of resources. Qualitatively, it helps utilize the expertise of judges and encourages knowledge sharing across a wider area [95]. Gupta & Bolia [96] propose optimization models to reallocate judges to enhance judicial output without any additional resources. He & Qu [97] propose judicial reallocations based on balancing workload per judge and staffing. Further, while recruiting judges for various positions, their experience should be given due consideration to ensure that they are a good fit for the designation.

5.5. Exogenous factors

The exogenous factors that are outside the ambit of the judiciary seem to be the least important factors. It can be said that the courts need to focus on structured and procedural reforms within the legal system to enhance performance and bring about a significant, long-term impact.

5.6. Analysis of interventions

Interventions should be developed and enforced after careful deliberations along with evidential support. Their applicability should be checked prior to implementation to ensure that they are suitable for the intended courts. Even though it may not be possible to empirically test the possible impact of all reforms, the effect of some reforms can be estimated based on different assumptions. For example, the effect of enhancing judicial strength on judicial output can be measured quantitatively based on historical data [91]. Once the reforms are put into effect, their impact must be regularly monitored to assess whether the intended benefits have been achieved or not. In case the actual improvements are not in line with expectations, the reforms can be appropriately modified to enable enhanced performance. For example, D'Amico & Manca [98] model the judicial system as a stochastic model and adopt a maintenance approach wherein the introduction of a reform is treated as a maintenance policy for the system.

6. Summary and conclusions

A country's judicial system should be capable of resolving cases without unreasonable delay while ensuring that due protocols are followed. The Indian judiciary is overburdened with litigation leading to slow processing of cases. There is an urgent need for effective reforms to enhance judicial output. While measures undertaken to tackle backlog by other judicial systems can be studied, similar rewards cannot be expected by their replication. Policy interventions need to be developed according to the factors affecting the performance of Indian courts; only then can the implementation of reforms yield desired gains.

In the current study, we develop a methodological framework to identify and prioritize factors that impact justice delivery and apply to the case of Indian courts. Twenty factors are determined through a review of literature and discussions with experts from the judicial space. Five broad categories are established to classify the factors for further analysis. The categories identified are organizational, judge-related, case-related, procedural and exogenous. A survey was administered to the experts to group the factors and the categorization was validated using Kappa statistic. Fuzzy best-worst method is applied to determine the overall ranks of factors affecting judicial performance. For this purpose, a survey was conducted with experts to determine the relative importance of broad categories and factors within the categories. Results indicate that among the broad categories, organizational and procedural are considered to be most important. The five factors with the highest ranks are: complexity associated with judicial procedures, lack of information technology initiatives, high complexity of cases, low judicial staff and procedural ambiguities. Based on the findings and discussions with respondents, policy recommendations are suggested.

Some of the key findings include: simplification of judicial processes through changes in procedural codes and acts are imperative to improve judicial performance. According to one of our survey experts, 'procedure has become a punishment in itself in Indian courts'. Judges and court staff can also adopt a more active role to improve processing of cases through the system. Complex cases need to be prioritised and can be subject to different procedures to reduce unnecessary delay. Existing applications of information technology need to be evaluated for usefulness and reforms in line with policy objectives need to implemented. Lastly, courts also need to enhance judicial staff through recruitments followed by their optimal allocations.

The study offers the first step for understanding the impact of multiple factors on judicial efficiency in India and can be extended in the following ways. The movement of cases through different stages in the court can be simulated to identify bottlenecks in the system and enable specific procedural changes. Separate surveys can be administered for different levels of courts. This will help determine the relative importance of similar factors across the judicial hierarchy. Similarly, surveys can be conducted across states to identify territorial variations in the influence of factors. The perceptions of litigants and law and order agencies can also be evaluated to develop an integrated framework for improvement in judicial performance.

Declaration of competing interest

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Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

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